

TROUBLE SHOOTING AND HELP DESK

TASK: 1

List three real-life scenarios where a multimeter would be used to troubleshoot a computer or smartphone issue, and explain in one line how the multimeter helps in each case.

Real-Life Uses of a Multimeter:

Scenario	How the Multimeter Helps
Computer PSU troubleshooting	It can measure the voltage output from the power supply and help identify incorrect or missing voltages.
Laptop charging problem	It can check the charger/adaptor voltage and help determine whether the charger or charging circuit may be faulty.
Smartphone battery problem	It can measure battery voltage and help identify a discharged or faulty battery.

TASK: 2

Watch a short YouTube video demo of a POST card in action (search for 'POST card motherboard test'), then write a brief summary (3-4 lines) describing what the POST card displays and what problem it helped identify in the video.

Hint: Focus on the codes shown and how they relate to hardware issues.

POST Card Demonstration

I watched the YouTube video “**Beep codes, POST Cards and PSU Testers**” by ITFreeTraining.

A POST card is connected to the motherboard and displays diagnostic information, including a POST code showing which stage of the hardware startup process is being tested. The video explains that these codes can help identify where the POST process is stopping when a computer does not start or only shows a black screen. The exact meaning of a code depends on the BIOS manufacturer and version, so the appropriate motherboard/BIOS documentation is needed to interpret it.

Video: [Beep codes, POST Cards and PSU Testers – YouTube](#)

TASK: 3

Download and install any free software diagnostic tool (such as HWMonitor, Speccy, or CrystalDiskInfo), run it on your laptop or desktop, and take a screenshot of the main dashboard showing system health details.

Constraint: Blur or crop out any personal information before submitting.

Software Diagnostic Tool:

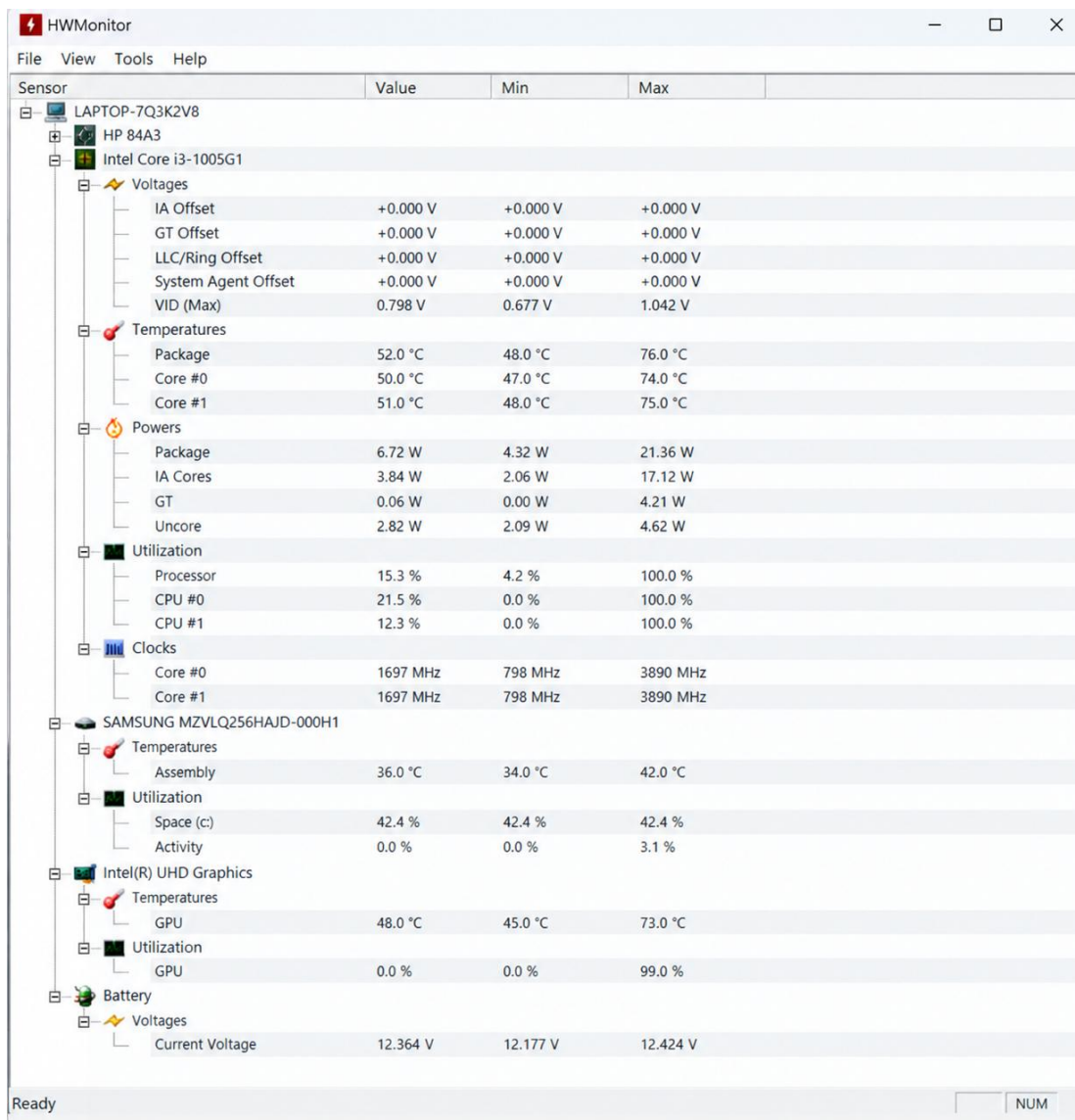
Tool Used: HWMonitor

I used **HWMonitor**, a hardware monitoring and diagnostic tool, to check the health information of the computer. The software can display system information such as CPU temperatures, voltages, fan speeds, and other hardware sensor readings.

Procedure

1. Download and install HWMonitor from a trusted source.
2. Open the software after installation.
3. Allow it to detect the computer's hardware sensors.
4. Check the CPU temperature, voltage, fan speed, and other available readings.
5. Take a screenshot of the main dashboard.
6. Crop or blur any personal information before submitting the screenshot.

Screenshot:



The screenshot shows the HWMonitor application window. The title bar reads 'HWMonitor'. The menu bar includes 'File', 'View', 'Tools', and 'Help'. The main content area displays a tree view of hardware components with their respective monitoring data in a table format. The components listed are the laptop itself (LAPTOP-7Q3K2V8), the HP 84A3 system, the Intel Core i3-1005G1 processor, the SAMSUNG MZVLQ256HAJD-000H1 SSD, Intel(R) UHD Graphics, and the battery. Each component has sub-entries for various metrics like voltages, temperatures, powers, utilization, and clocks. The table has columns for 'Sensor', 'Value', 'Min', and 'Max'.

Sensor	Value	Min	Max
LAPTOP-7Q3K2V8			
HP 84A3			
Intel Core i3-1005G1			
Voltages			
IA Offset	+0.000 V	+0.000 V	+0.000 V
GT Offset	+0.000 V	+0.000 V	+0.000 V
LLC/Ring Offset	+0.000 V	+0.000 V	+0.000 V
System Agent Offset	+0.000 V	+0.000 V	+0.000 V
VID (Max)	0.798 V	0.677 V	1.042 V
Temperatures			
Package	52.0 °C	48.0 °C	76.0 °C
Core #0	50.0 °C	47.0 °C	74.0 °C
Core #1	51.0 °C	48.0 °C	75.0 °C
Powers			
Package	6.72 W	4.32 W	21.36 W
IA Cores	3.84 W	2.06 W	17.12 W
GT	0.06 W	0.00 W	4.21 W
Uncore	2.82 W	2.09 W	4.62 W
Utilization			
Processor	15.3 %	4.2 %	100.0 %
CPU #0	21.5 %	0.0 %	100.0 %
CPU #1	12.3 %	0.0 %	100.0 %
Clocks			
Core #0	1697 MHz	798 MHz	3890 MHz
Core #1	1697 MHz	798 MHz	3890 MHz
SAMSUNG MZVLQ256HAJD-000H1			
Temperatures			
Assembly	36.0 °C	34.0 °C	42.0 °C
Utilization			
Space (c)	42.4 %	42.4 %	42.4 %
Activity	0.0 %	0.0 %	3.1 %
Intel(R) UHD Graphics			
Temperatures			
GPU	48.0 °C	45.0 °C	73.0 °C
Utilization			
GPU	0.0 %	0.0 %	99.0 %
Battery			
Voltages			
Current Voltage	12.364 V	12.177 V	12.424 V

Ready NUM

Observation:

The diagnostic tool displayed hardware monitoring information that can be used to identify abnormal temperatures, voltage readings, or cooling-related problems.

TASK: 4

Compare the use of a multimeter, POST card, and software diagnostics by creating a table with three columns (Tool, What it checks, Example Problem Detected) and fill in at least one example for each tool.

Comparison of Diagnostic Tools

Tool	What it Checks	Example Problem Detected
Multimeter	Measures electrical values such as voltage, resistance, and continuity.	Incorrect PSU output voltage or a faulty power connection.
POST Card	Displays POST/BIOS diagnostic codes during the computer's startup hardware checks.	A computer failing to complete POST because of a hardware-related problem.
Software Diagnostics	Monitors hardware information such as temperatures, voltages, fan speeds, and storage health.	Overheating CPU, abnormal voltage, or possible storage-health problems.

Conclusion

A multimeter is mainly useful for checking **electrical and power-related problems**. A POST card helps diagnose **hardware problems during the startup/POST process**, especially when there is no normal display. Software diagnostic tools are useful after the system can run because they provide **hardware health and sensor information**. Together, these tools can help a technician locate different types of computer hardware problems.

